Reading Report: DRB-GAN and AdaIN

Image Style Transfer

Image style transfer aims to give a photograph the artistic appearance of a reference painting while keeping its main content recognisable. For example, a campus photograph could take on the colours and brushstrokes of an oil painting while retaining the buildings and their layout. I read DRB-GAN as a GAN-based approach to this task and AdaIN as a simpler method that helps explain one of its underlying techniques.

DRB-GAN: Learning from Individual Paintings and Collections

DRB-GAN, presented by Xu et al. at ICCV 2021, considers both arbitrary style transfer and collection style transfer. The first uses an individual reference image. The second draws on a group of paintings to represent the shared style of an artist or collection. This distinction interested me because one painting may contain unusual features that do not represent an artist's work as a whole. [1, Introduction]

The model first extracts a style code from the reference artwork. I understand this code as a set of instructions for changing how the content image is processed. Its Dynamic ResBlocks combine dynamic convolution and AdaIN, allowing some parameters to change with the style input. During training, a generator produces stylised images, while a discriminator uses paintings from the target collection as references. Adversarial training encourages the generated images to match the target style. Perceptual supervision, including a content loss, also helps preserve the original scene. [1, Sections 3.1-3.4]

What I found useful was the connection between individual examples and a broader artistic style. The authors report results for both kinds of transfer. For a course project, however, I would also need to consider how to organise the painting collections and train the different components together. Attractive examples alone would not tell me how well the method works on our own images. [1, Section 4]

AdaIN: A Simple Way to Control Style

Huang and Belongie's AdaIN paper, presented at ICCV 2017, addresses a trade-off in earlier methods. Some could transfer different styles but required slow optimisation for each output. Faster methods were often restricted to styles chosen during training. AdaIN aims to let one trained network quickly process new reference styles. [2, Introduction]

The method uses a VGG encoder to extract features from the content and style images. AdaIN then aligns the channel-wise means and standard deviations of the content features with those of the style features. A trained decoder converts the adjusted features back into an image. The operation changes statistics in feature space, rather than directly replacing the average colour of the original pixels. [2, Sections 5-6]

I found the simplicity of this idea appealing. A small, understandable operation can provide useful control over style without a complicated transformation procedure. However, these statistics cannot fully describe every brushstroke or texture. The authors also discuss a trade-off between speed, flexibility and visual quality. AdaIN is therefore a useful starting point and baseline, rather than a complete answer to every style transfer problem. [2, Sections 7-8]

How the Two Papers Connect

AdaIN is not itself a GAN method. Its original network contains an encoder, an AdaIN layer and a decoder, and is trained using content and style losses without a discriminator. DRB-GAN directly uses AdaIN layers inside its Dynamic ResBlocks. Its style encoding network predicts the corresponding affine parameters, rather than copying the complete network from the original AdaIN paper. The two readings therefore connect a full GAN-based method with one of its underlying techniques. [1, Sections 3.1-3.2; 2, Section 6]

Personal Reflection

These papers changed how I think about style transfer. I initially thought of it as applying an artistic filter to a photograph. I now see separate questions about how style is represented, how it controls generation, and how the original content is preserved. AdaIN makes the first two questions easier to understand. DRB-GAN shows how style control can work alongside a discriminator and additional training objectives.

For our project, I would use DRB-GAN as a main reference for the GAN approach and consider AdaIN as a baseline. I would examine whether stronger stylisation also removes important details, such as a building's edges or a person's outline. I would also compare processing speed and behaviour at different resolutions. Using a GAN does not by itself establish that global structure is preserved. These properties would still need to be evaluated on the images used in our project.

References

[1] Xu, W., Long, C., Wang, R., and Wang, G. (2021). *DRB-GAN: A Dynamic ResBlock Generative Adversarial Network for Artistic Style Transfer*. Proceedings of ICCV, pp. 6383-6392. https://xuwenju123.github.io/publications/DRBGAN_ICCV.pdf

[2] Huang, X., and Belongie, S. (2017). *Arbitrary Style Transfer in Real-Time With Adaptive Instance Normalization*. Proceedings of ICCV. https://arxiv.org/html/1703.06868v2